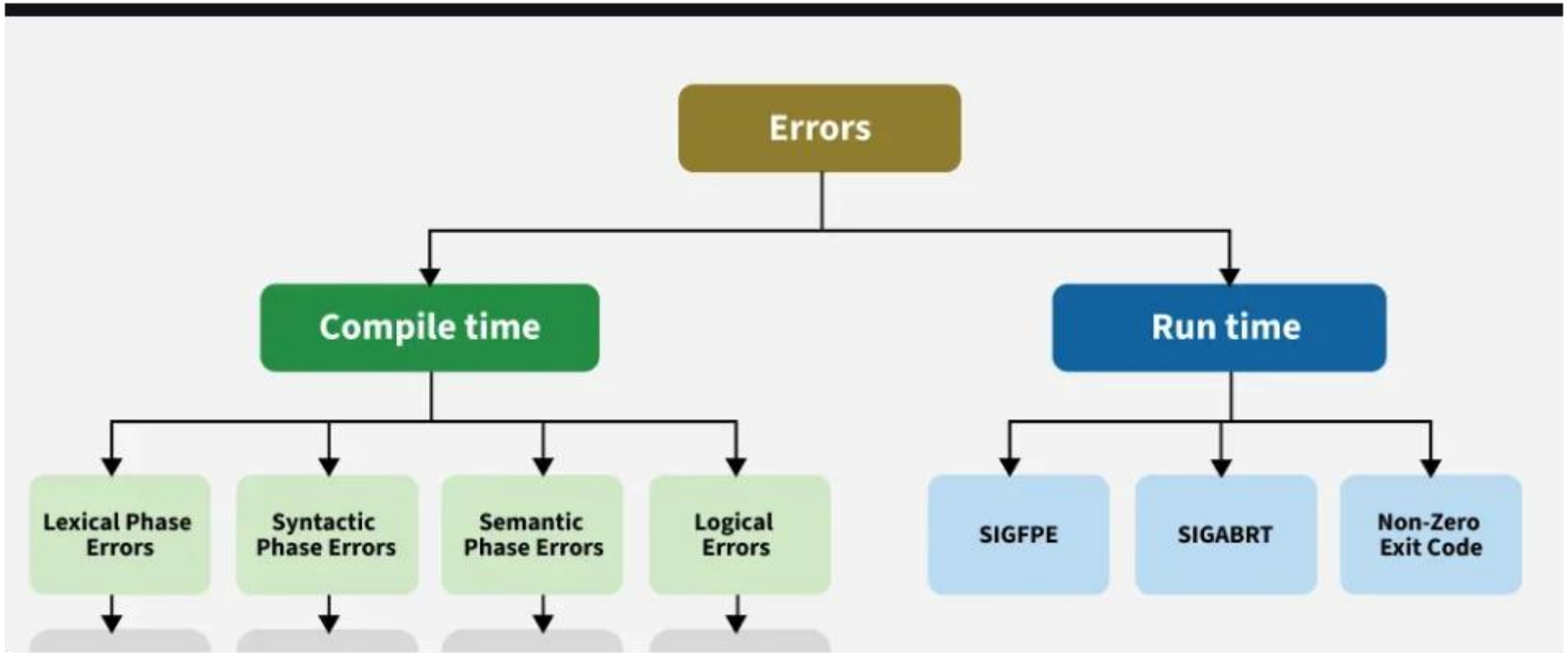
Error handling in compiler design

Errors are mistakes in a program that cause incorrect results or prevent the program from compiling or executing correctly.

During the process of language translation, the compiler can encounter errors. While the compiler might not always know the exact cause of the error, it can detect and analyze the visible problems. The main purpose of error handling is to assist the programmer by pointing out issues in their code.

Error handler = Error Detection + Error Report + Error Recovery.

- **Sources of Error in Error Handling**
- An Error is the blank entries in the symbol table. Errors in the program should be detected and reported by the parser.
- Whenever an error occurs, the parser can handle it and continue to parse the rest of the input.
- Although the parser is mostly responsible for checking for errors, errors may occur at various stages of the compilation process.
- Error handling is a process that helps identify errors in a program, reports them to the user, and then applies recovery strategies to manage the errors. During this process, it's important that the program's processing time is not slowed down too much. One common error source is blank entries in the symbol table.


Errors
Compile time
Run time
Lexical Phase Errors
Syntactic Phase Errors
Semantic Phase Errors
Logical Errors
SIGFPE
SIGABRT
Non-Zero Exit Code

- **1. Run-Time Errors**
- These errors happen while the program is running. They usually occur because of issues like incorrect system settings or invalid input data. Examples of run-time errors include:
- **Lack of memory** to run the program.
- **Memory conflicts** with other programs.
- **Logical errors**, where the program doesn’t produce the expected results. These can be fixed by carefully debugging the code.

- **2. Compile-Time Errors**
- These errors occur before the program starts running, during the compilation process. They stop the program from compiling successfully.
  Examples of compile-time include:
- **Syntax errors** (like missing semicolons or incorrect statements).
- **Missing file references** that prevent the program from compiling.
-

2. Syntactic phase errors

These errors are detected during the syntax analysis phase. Typical syntax errors are:

- Errors in structure
- Missing operator
- Misspelled keywords
- Unbalanced parenthesis

```
Example : swich(ch)
          {
            .......
            .......
          }
```

- Compile-time errors are of three types:-
- **. Lexical phase errors**
- These errors are detected during the lexical analysis phase. Typical lexical errors are:
- Exceeding length of identifier or numeric constants.
- The appearance of illegal characters
- Unmatched string

3. Semantic errors

These errors are detected during the semantic analysis phase. Typical semantic errors are :

- Incompatible type of operands
- Undeclared variables
- Not matching of actual arguments with a formal one

```
Example : int a[10], b;
          .......
          .......
          a = b;
```

It generates a semantic error because of an incompatible type of a and b.